



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Lipid metabolism Part II: metabolism of lipoproteins

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**Harvey and Ferrier: Lippincott Biochemistry 8th Edition: Chapter 15 and 18
Marks' Basic Medical Biochemistry, 6e, Chapter 32 and 34**

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Learning Objectives

1. Explain the digestion and absorption of dietary lipids
2. Illustrate the function of the chylomicron pathway
3. Explain the causes and consequences of fat malabsorption
4. Compare and contrast the life cycle of the various lipoprotein particles (VLDL, IDL, LDL, and HDL) with respect to their functions, composition, metabolism and transport.
5. Describe the clinical relevance related to dietary lipid absorption

Structure of lipoprotein particles

The structure of plasma lipoproteins - I

- Both **triglyceride** (or triacylglycerol) and **cholesterol** may be absorbed from diet or synthesized in the body (de novo). They cannot be **transported** as free molecules due to their nonpolar nature
- Triglyceride and cholesterol are transported in plasma as **lipoproteins** particles— complexes of lipids and proteins (apoproteins). Lipoproteins are mixed **micelles** consisting of a core of nonpolar triglycerides and cholesterol (esters) surrounded by a surface layer of phospholipids and apoproteins.

Triglyceride (TG) or
Triacylglycerol (TAG)

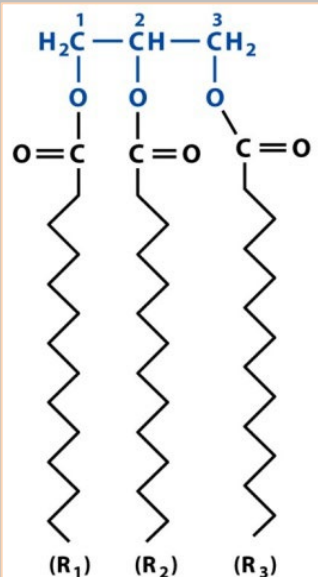
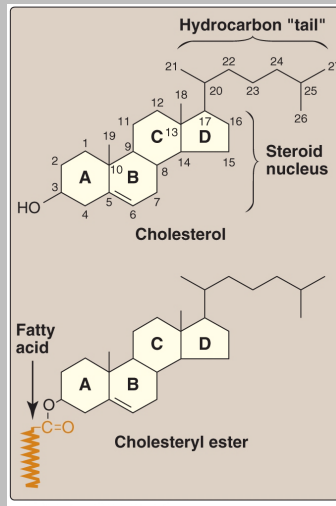
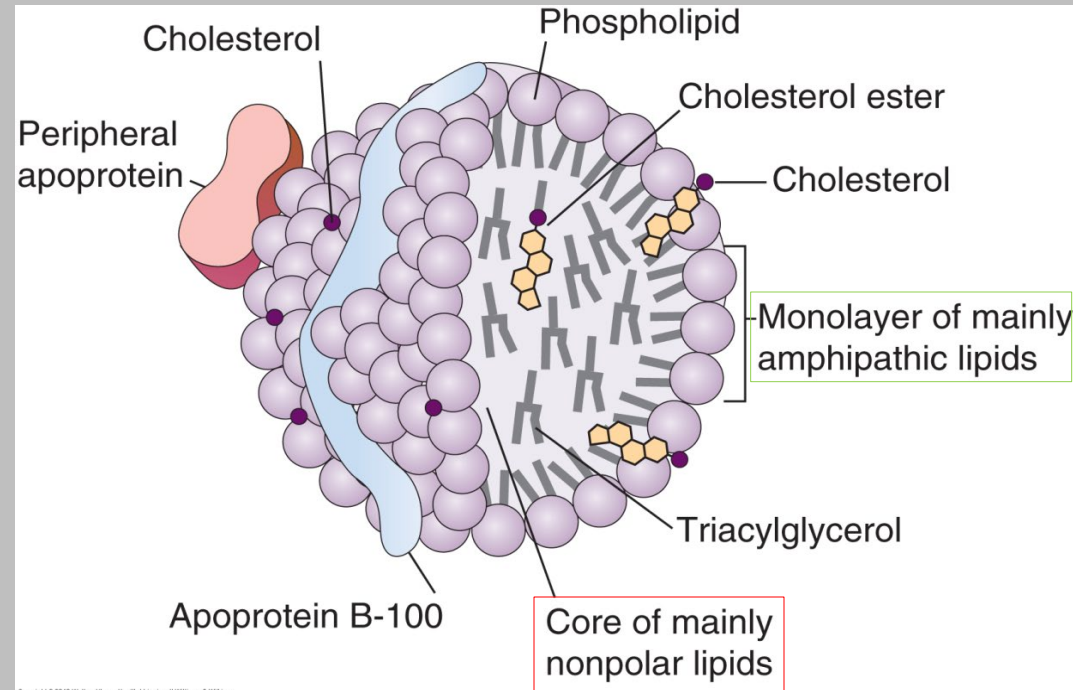


Figure 9-5b Principles of Biochemistry, 4/e
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Cholesterol

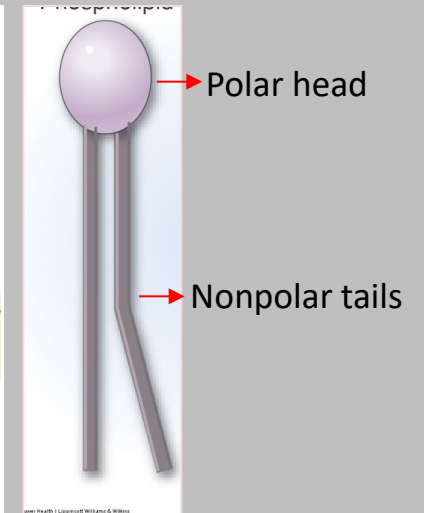


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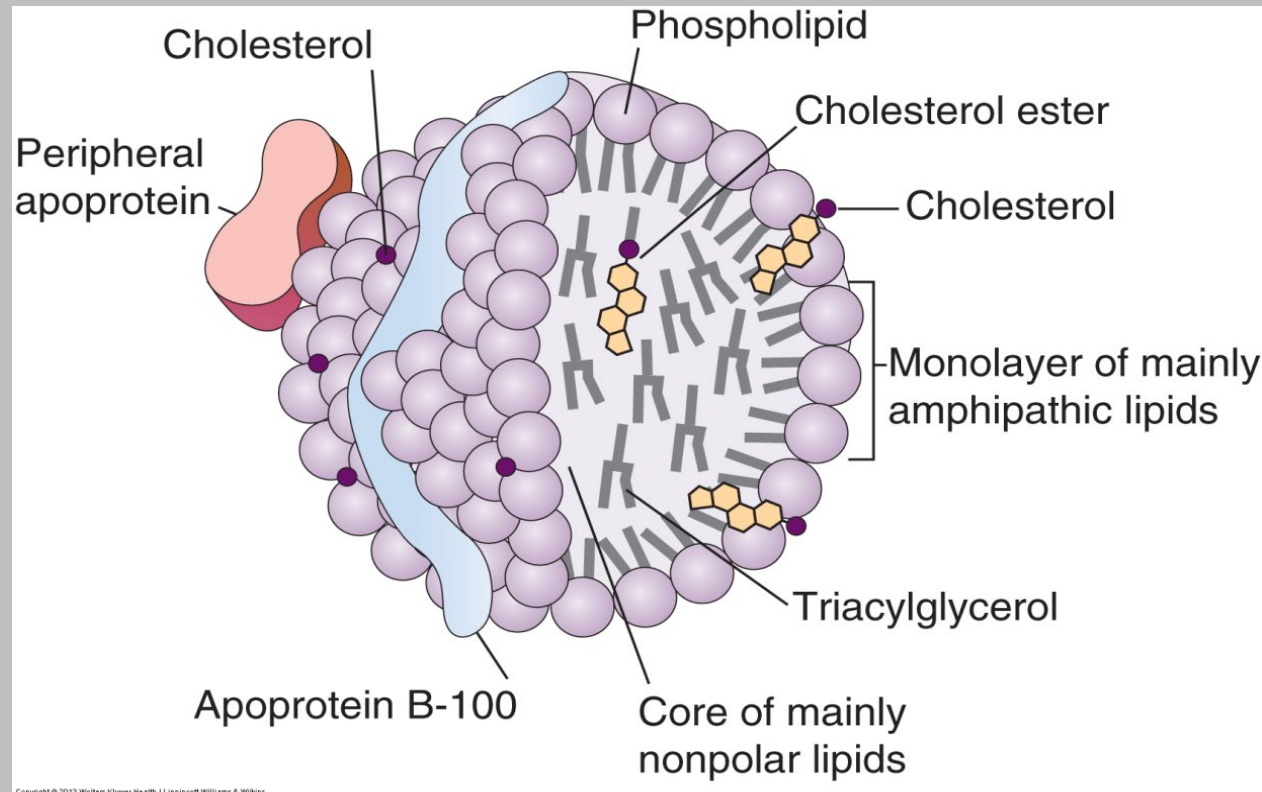
Phospholipid



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The structure of plasma lipoproteins - II

- Triglycerides are low density. Proteins are high density. So, the more triglycerides in a lipoprotein particle, the lower its density.
- Lipoproteins can be separated from each other by electrophoresis or density gradient centrifugation.



A summary of the functions of lipoproteins

Functions:

- “**Solubilize**” the insoluble lipids for transport in the blood
- **Transport** both dietary lipids and endogenously synthesized lipids

Classification:

- **Chylomicrons** transport **dietary lipids** and **fat-soluble vitamins** from intestine to liver and other tissues
- **VLDLs** transport **endogenous synthesized lipids** from liver to other tissues
- **LDLs** deliver **cholesterol** from liver to tissues with more cholesterol needs (adrenals, sex glands, etc.)
- **HDLs** transport **cholesterol** from peripheral tissues back to the liver

Major classes of lipoproteins

The more triglyceride in a lipoprotein particle, the lower its density:

- Chylomicrons (the **least dense**)
- Very densest lipoproteins (VLDL)
- Intermediate density lipoproteins (IDL)
- Low density lipoproteins (LDL)
- High density lipoproteins (HDL) (the **densest**)

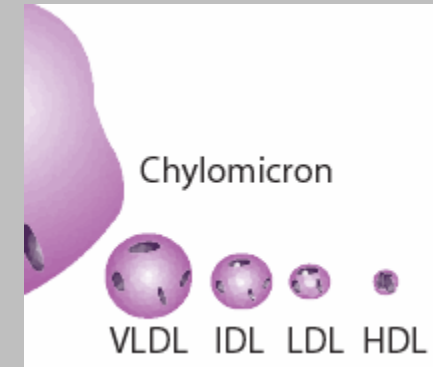


TABLE 16.1 Lipoproteins in human plasma

	Chylomicrons	VLDLs	IDLs	LDLs	HDLs
Molecular weight $\times 10^{-6}$	$\leftarrow >400$	10–80	5–10	2.3	0.18–0.36
Density (g cm^{-3})	<0.95	0.95–1.006	1.006–1.019	1.019–1.063	1.063–1.210 $\rightarrow$
Chemical composition (%)					
Protein	2	10	18	25	33 $\rightarrow$
Triacylglycerol	$\leftarrow 85$	50	31	10	8
Cholesterol	4	22	29	45	30
Phospholipid	9	18	22	20	29

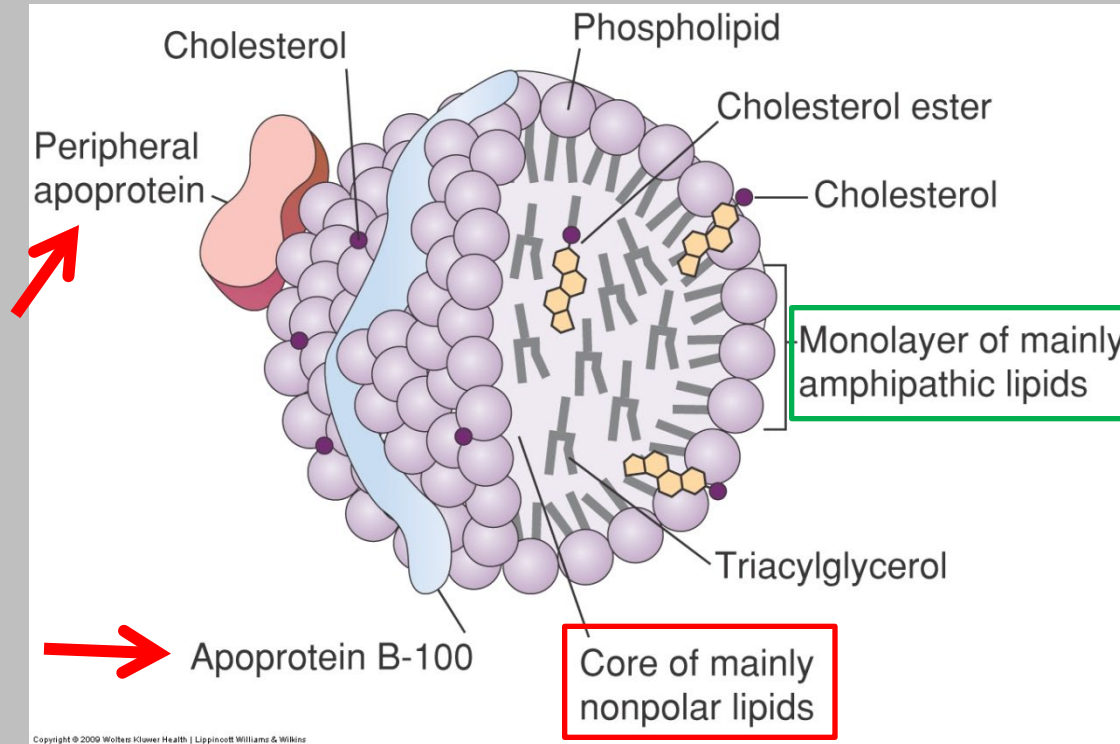
Characteristics of the Major Lipoproteins

Lipoprotein	Density Range (g/mL)	Particle Diameter (nm) Range	Electrophoretic Mobility	Lipid (%) ^a			Function
				TG	Chol	PL	
Chylomicrons	0.930	75–1200	Origin	80–95	2–7	3–9	Deliver dietary lipids
Chylomicron remnants	0.930–1.006	30–80	Slow pre- β				Return dietary lipids to the liver
VLDL	0.930–1.006	30–80	Pre- β	55–80	5–15	10–20	Deliver endogenous lipids
IDL	1.006–1.019	25–35	Slow pre- β	20–50	20–40	15–25	Return endogenous lipids to the liver; precursor of LDL
LDL	1.019–1.063	18–25	β	5–15	40–50	20–25	Deliver cholesterol to cells
HDL ₂	1.063–1.125	9–12	α	5–10	15–25	20–30	Reverse cholesterol transport
HDL ₃	1.125–1.210	5–9	α				Reverse cholesterol transport
Lip(a)	1.050–1.120	25	Pre- β				

TG, triacylglycerol; Chol, the sum of free and esterified cholesterol; PL, phospholipid; VLDL, very low-density lipoprotein; IDL, intermediate-density lipoprotein; LDL, low-density lipoprotein; HDL, high-density lipoprotein.

^aThe remaining percentage composition is composed of apoproteins.

Structural features of plasma lipoproteins -apoproteins



Chylomicron
VLDL
IDL
LDL
HDL

- **Shell:** phospholipids and other amphipathic molecules on the surface
- **Core** contains lipids (triglyceride and cholesterol esters)
- **Apoproteins** (also amphipathic):
 - Act as structural components (e.g., apoB-48, apoB-100, apoA)
 - Regulate enzyme activity (e.g., apoA-I, apoC-II)
 - Act as ligands for receptors (e.g., apoB, apoE)

Characteristics of the Major Apolipoproteins

APOLIPOPROTEIN	PRIMARY TISSUE SOURCE	MOLECULAR MASS (DALTONS)	LIPOPROTEIN DISTRIBUTION	METABOLIC FUNCTION
→ ApoA-I	Intestine, liver	28,016	HDL (chylomicrons)	Activates LCAT; structural component of HDL
ApoA-II	Liver	17,414	HDL (chylomicrons)	Structural proteins for HDL; activates hepatic lipase
ApoA-IV	Intestine	46,465	HDL (chylomicrons)	Uncertain; may be involved in assembly of HDL and chylomicrons
ApoA-V	Liver	39,000	VLDL, chylomicrons, HDL	Promotes LPL mediated triglyceride hydrolysis
→ ApoB-48	Intestine	241,000	Chylomicrons	Assembly and secretion of chylomicrons from small bowel
→ ApoB-100	Liver	512,000	VLDL, IDL, LDL, Lp(a)	VLDL assembly and secretion; structural protein of VLDL, IDL, and LDL; ligand for LDL receptor
→ ApoC-I	Liver	6,630	Chylomicrons, VLDL, HDL	Activates LCAT
→ ApoC-II	Liver	8,800	Chylomicrons, VLDL, HDL	Cofactor activator of lipoprotein lipase (LPL)
ApoC-III	Liver	8,800	Chylomicrons, VLDL, HDL	Inhibitor of LPL; inhibits hepatic uptake of chylomicrons and VLDL remnants
→ ApoE	Liver	34,145	Chylomicron remnants, IDL, HDL	Ligand for binding of several lipoproteins to the LDL receptor; to the LDL receptor-related protein (LRP) and possibly to a separate apoE receptor
→ Apo(a)	Liver	250,000–800,000	Lipoprotein “little” a [Lp(a)]	Consists of apoB100 linked by a disulfide bond to apolipoprotein (a); inhibits plasminogen activation and fibrinolysis

HDL, high-density lipoprotein; IDL, intermediate-density lipoprotein; LCAT, lecithin-cholesterol acyltransferase; LDL, low-density lipoprotein; VLDL, very-low-density lipoprotein.

Adapted from Feingold KR. Introduction to lipids and lipoproteins. *Endotext* [Internet]. <https://www.ncbi.nlm.nih.gov/books/NBK305896/>

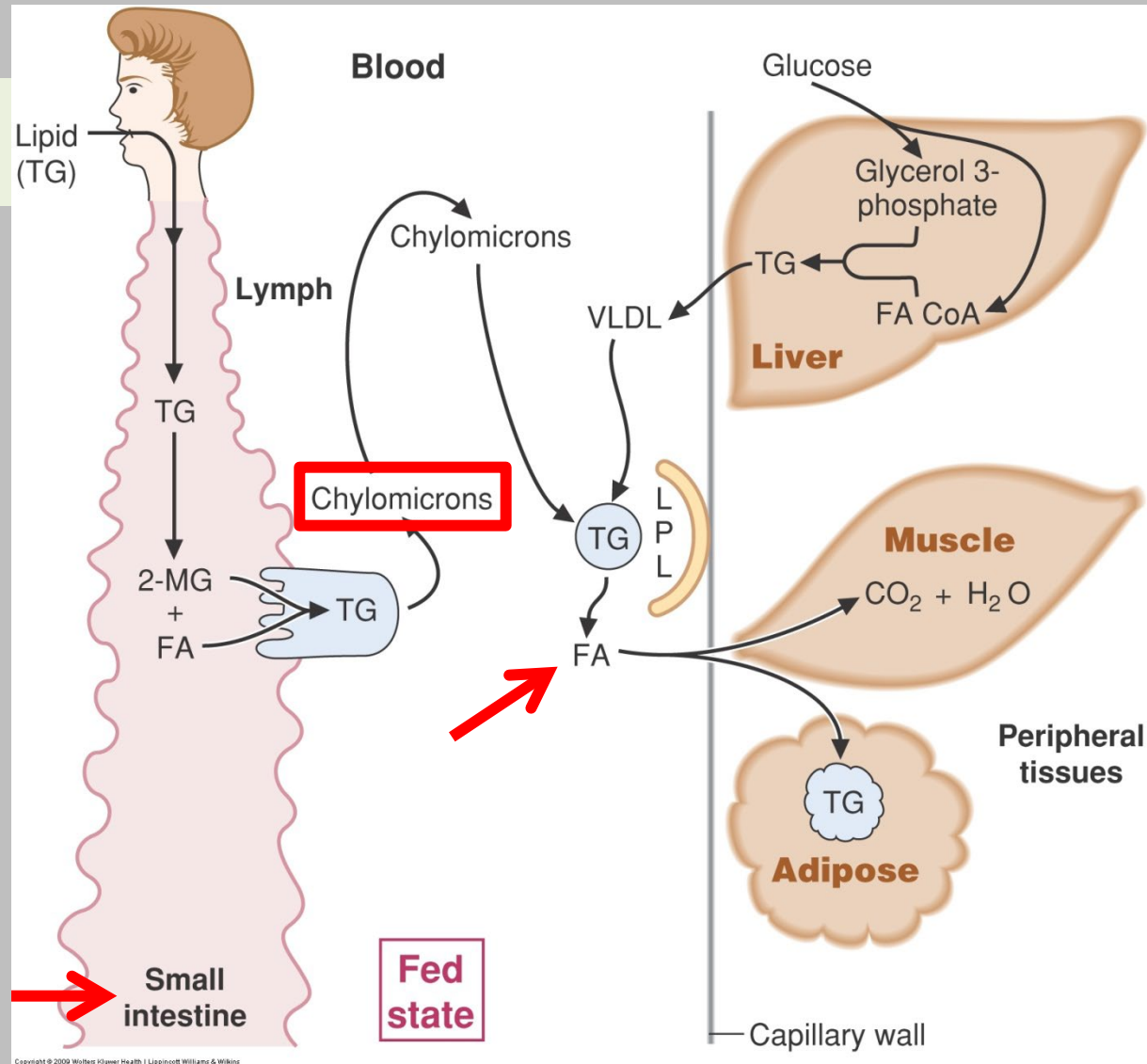
The three major pathways of lipoprotein metabolism

- The **chylomicron/chylomicron remnant pathway**: distributes **dietary triglycerides** and **cholesterol** to **liver** and peripheral tissues
- The **very low density lipoproteins (VLDL)/intermediate density lipoproteins (IDL)/low density lipoproteins (LDL) pathway**: transports **triglycerides** and **cholesterol** out of **liver** to the peripheral tissues
- The **nascent/mature high density lipoprotein (HDL) pathway**: transports **cholesterol** from peripheral tissues back to **liver**

Transport of dietary lipids - Chylomicrons

Absorption and transport of dietary lipids - overview

**Dietary
(exogenous)**

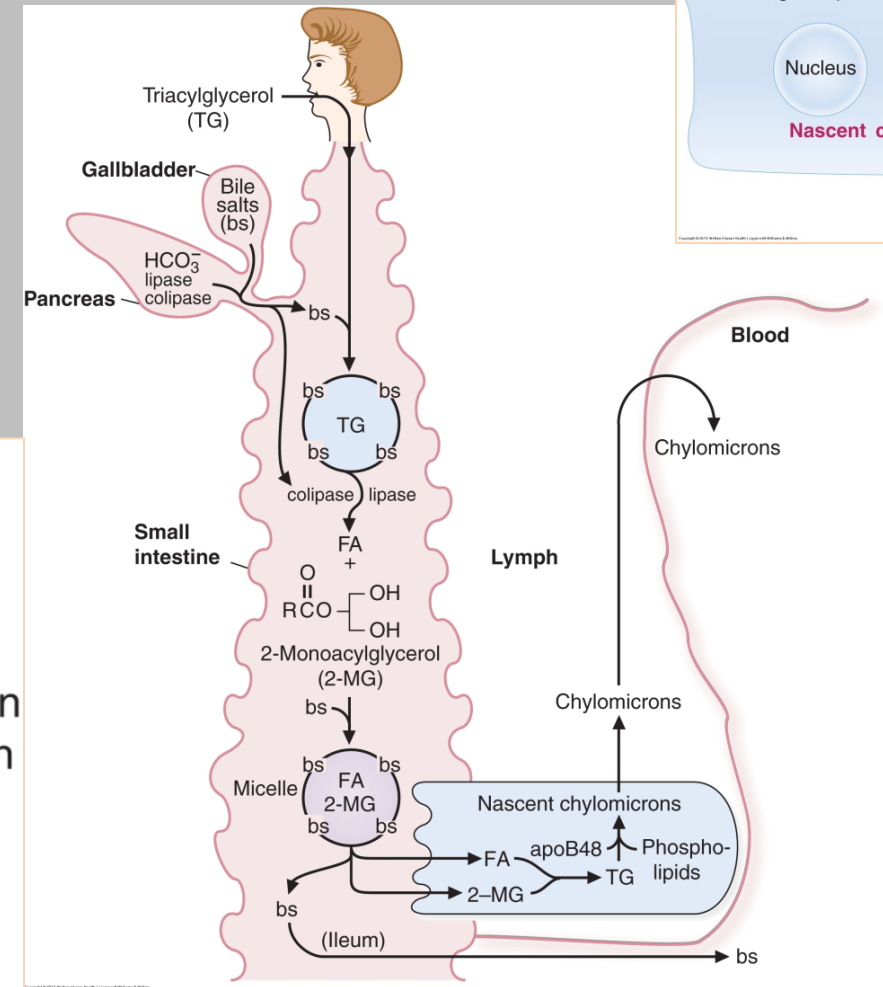
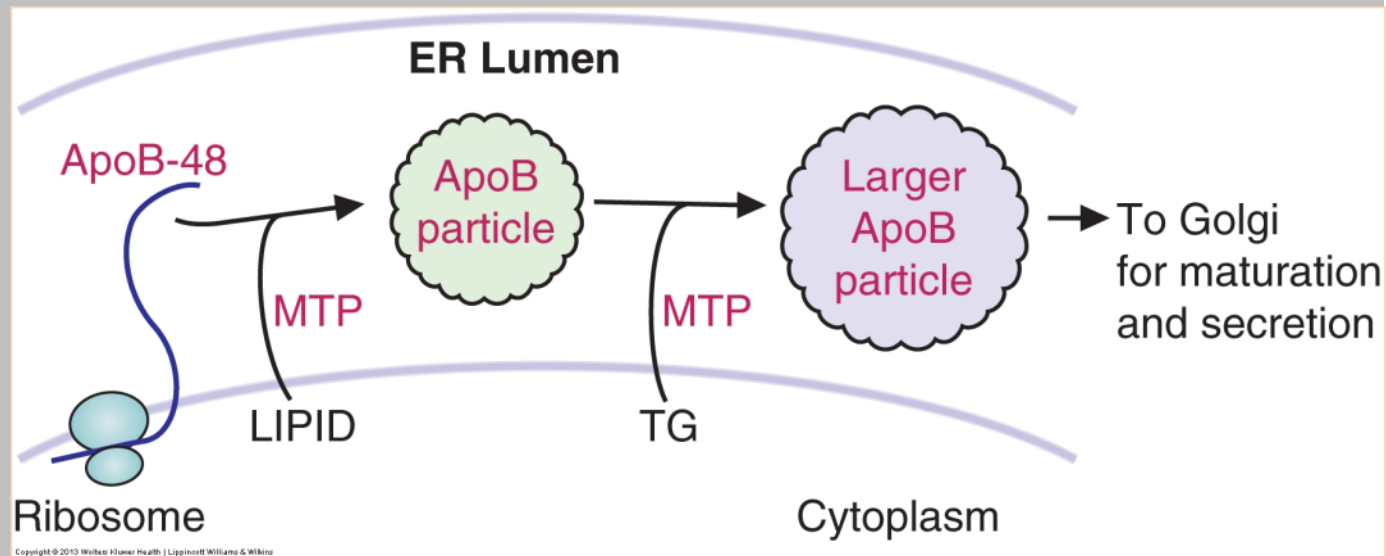
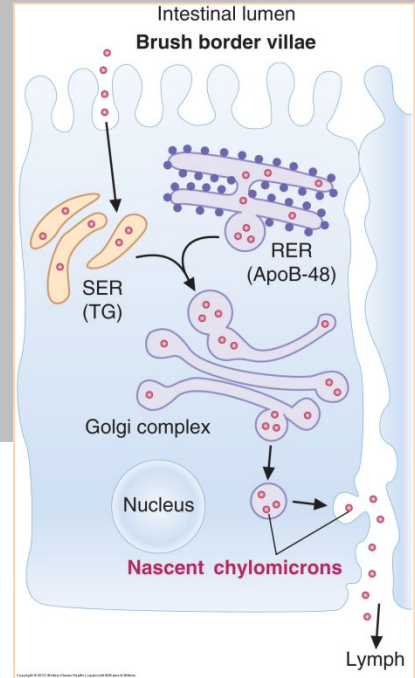


TG: triglyceride
2-MG: 2-monoacylglycerol
FA: fatty acid
LPL: lipoprotein lipase

Transport of dietary lipids by chylomicrons - synthesis

Chylomicrons: synthesized in the **intestinal epithelial cells**:

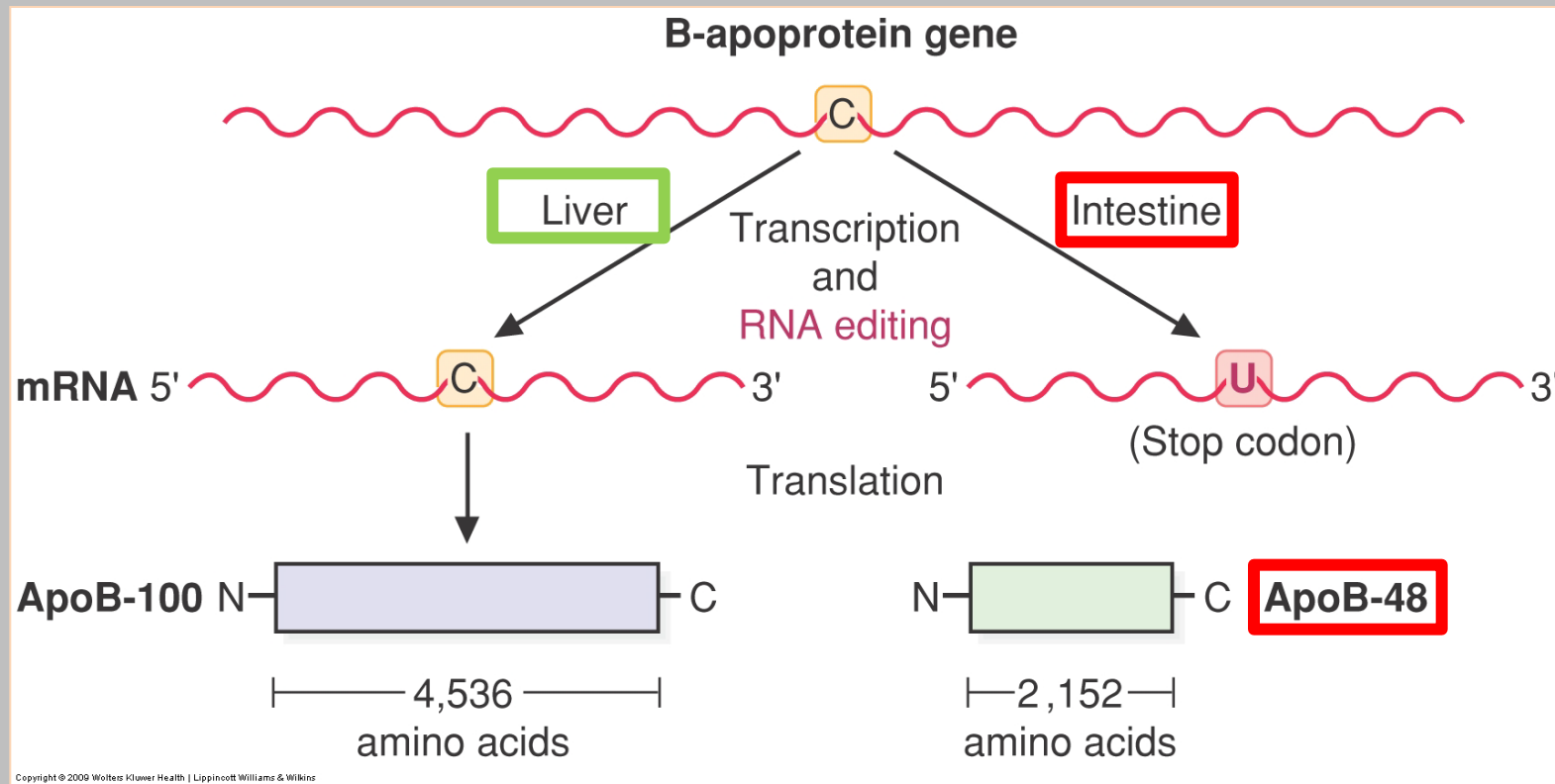
- **ApoB-48** is produced in ER (next slide).
- Lipids are loaded onto **ApoB-48** by **MTP** (or MTTP, **Microsomal triglyceride Transfer Protein**).
- Defect in MTP causes **Abetalipoproteinemia**
- Secreted into the **lymphatic system**
- Enter the blood through the **thoracic duct**



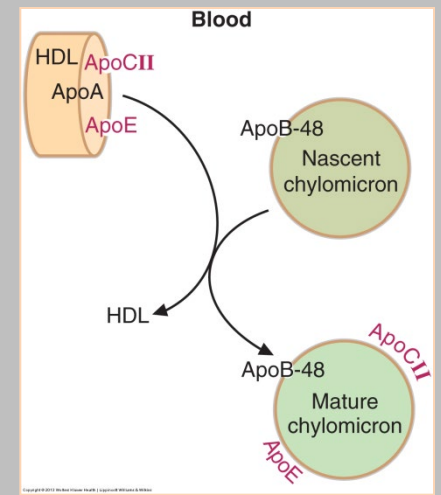
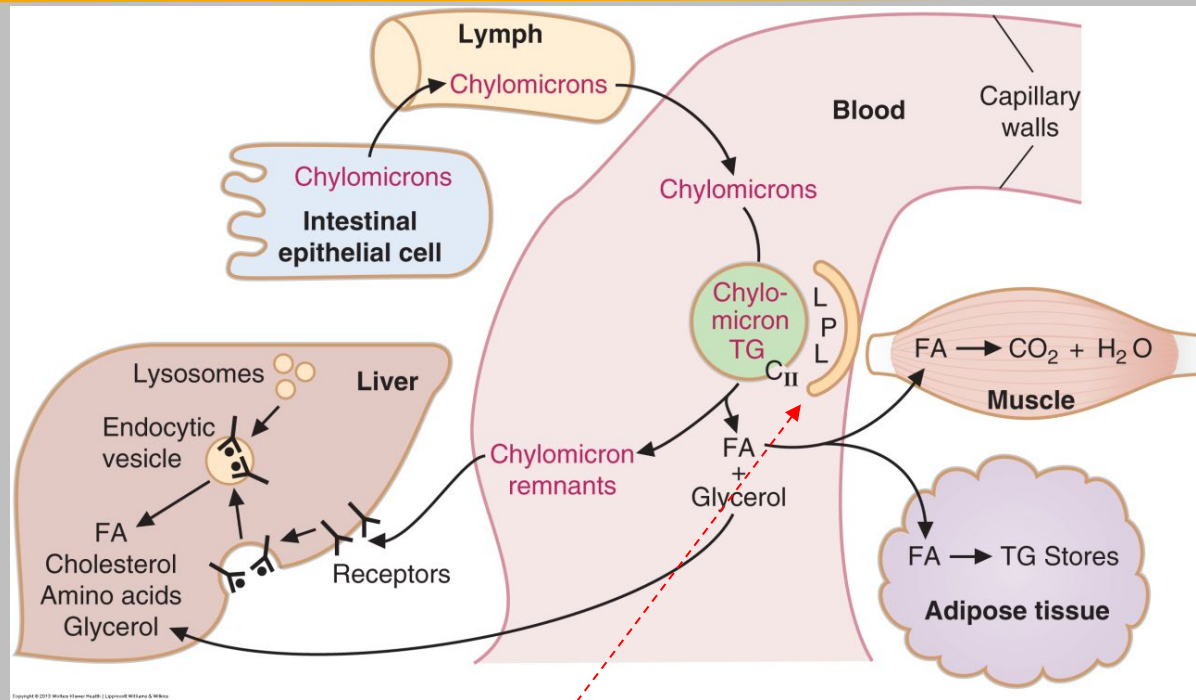
Transport of dietary lipids by chylomicrons – ApoB-48

ApoB-48 is produced by RNA editing of the ApoB-100 gene

- RNA editing takes place in the intestine
- C edited to U to create a **STOP** codon



Transport of dietary lipids by chylomicrons: maturation and fate



Maturation - transfer of apoproteins from HDLs to the nascent chylomicrons:

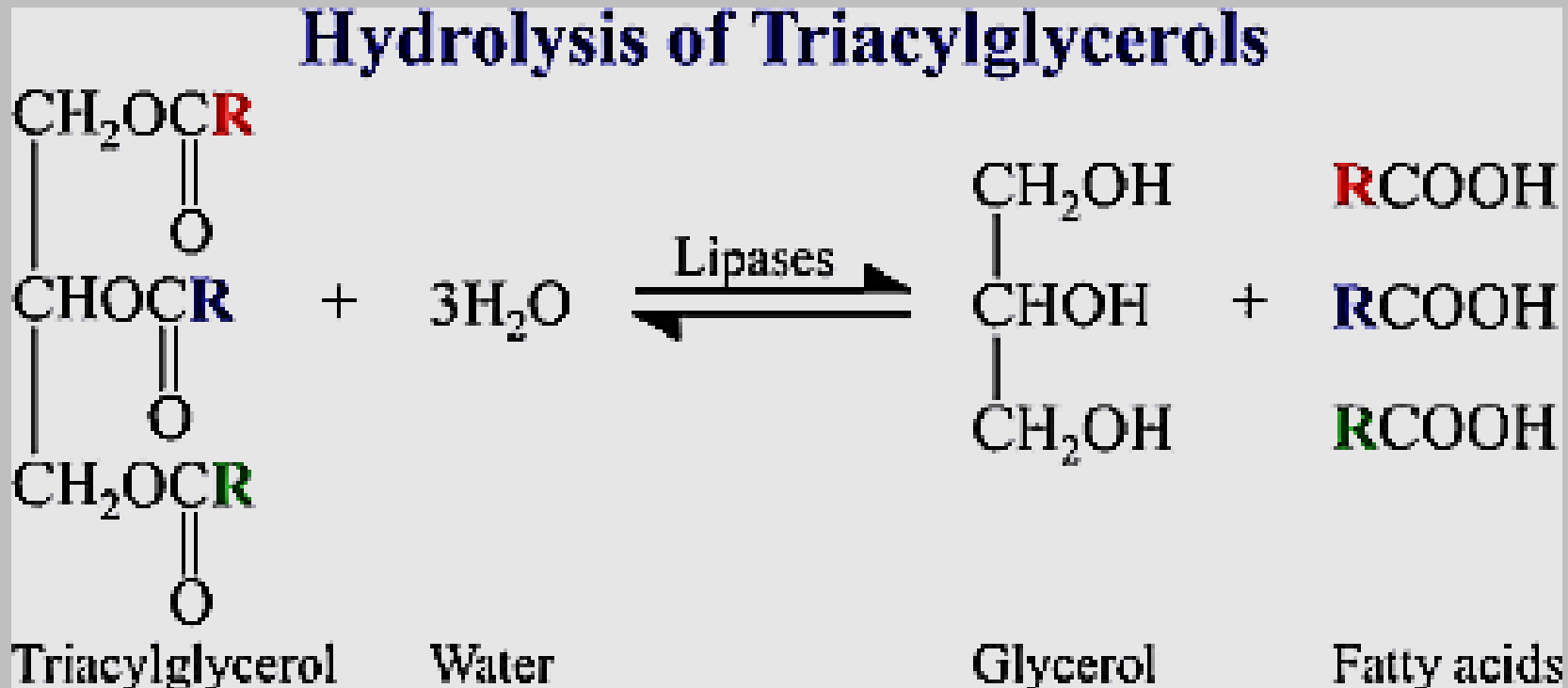
- **ApoC-II** and **ApoE** are transferred to the nascent chylomicrons to form mature chylomicrons
 - **ApoC-II** stimulates LPL
 - **ApoE** recognized by receptors

Fate - **Lipoprotein lipase (LPL)** hydrolyzes the TGs in the chylomicrons:

- LPL is attached to proteoglycans on the **basement membrane** of capillary endothelium
- LPL is high in adipose tissues, muscle, and mammary gland
 - **Insulin** stimulates LPL synthesis in adipose tissues after a balanced meal
- LPL is activated by **ApoC-II**
 - Released fatty acids may enter the cell or bind albumin in blood
 - glycerol used by the liver for lipid synthesis, glycolysis, or gluconeogenesis
- The remaining chylomicron is thus called **chylomicron remnant**, which can be taken up by liver through binding to the receptors on the surface of hepatocytes and endocytosis

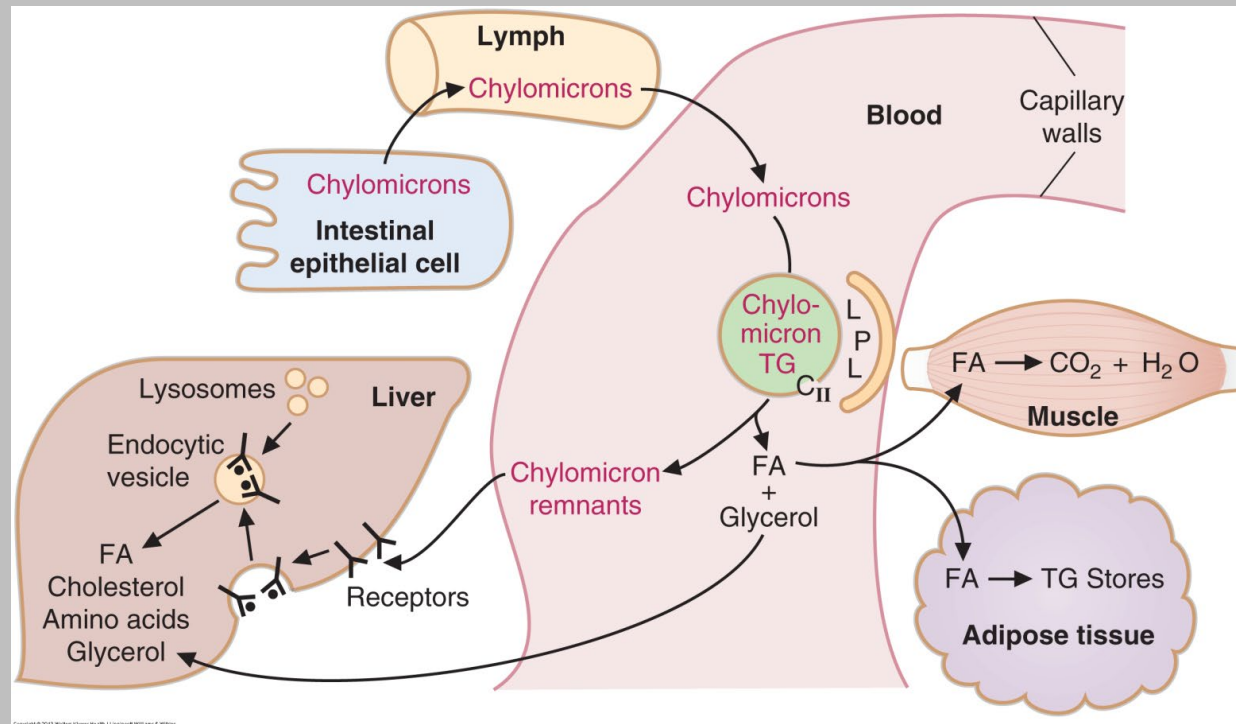
The activity of lipoprotein lipase (LPL)

Fatty acids are removed from triglycerides one at a time by **LPL**



Transport of dietary lipids by Chylomicrons - fate

- **Chylomicron remnants** decrease in size and increase in density after lose of TGs
- **ApoC-II** returns to **HDLs**
- **Chylomicron remnants** can be taken up by the hepatocytes
 - Bound to the hepatocyte **apoE receptor** via **apoE**
 - Taken into hepatocytes through endocytosis
 - The endocytic vesicles subsequently fuse with lysosomes
 - The contents are hydrolyzed and used by liver for biosynthesis





Clinical relevance to the absorption of dietary lipids

Fat malabsorption

- Possible causes:
 - Pancreatic failure (pancreas secretes a variety of **digestive enzymes** including lipase and colipase**)
 - Lack of **bile salts** (bile duct blockage)
 - Intestinal diseases
- Malabsorption causes **Steatorrhea** with bulky fatty appearing in stools
- Decrease in **fat soluble vitamin** absorption (D,E,A,K) because they are absorbed and transported via the Chylomicron pathway as well.

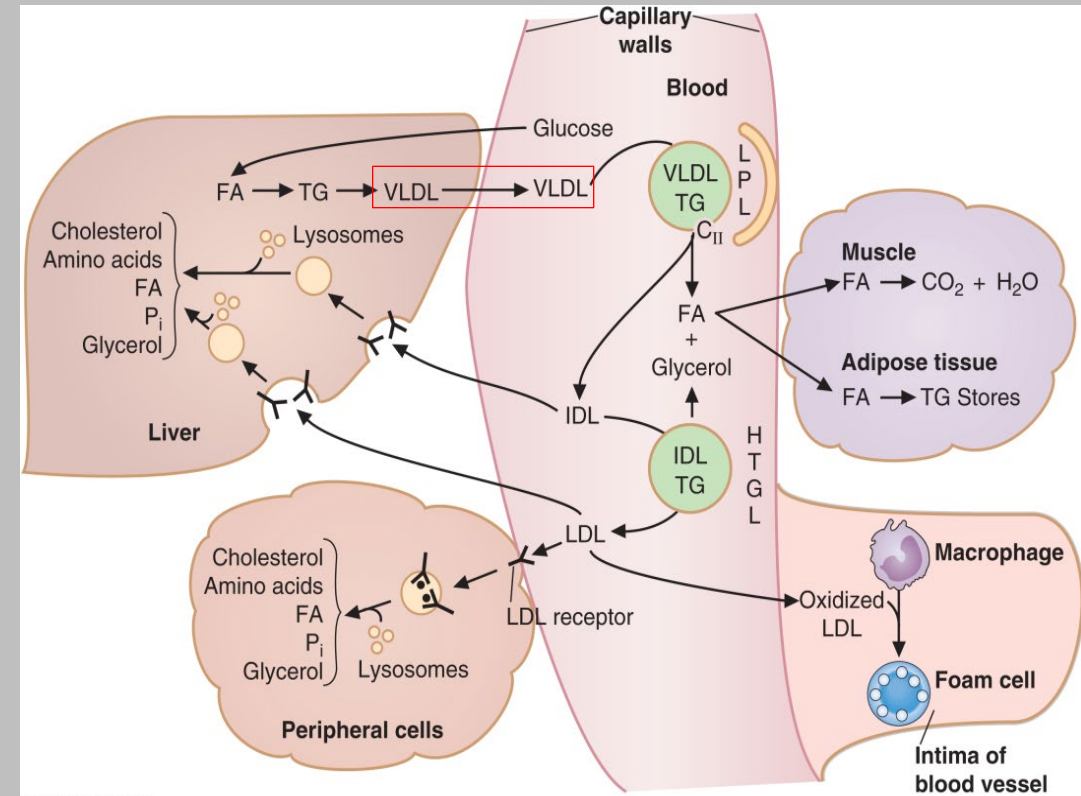
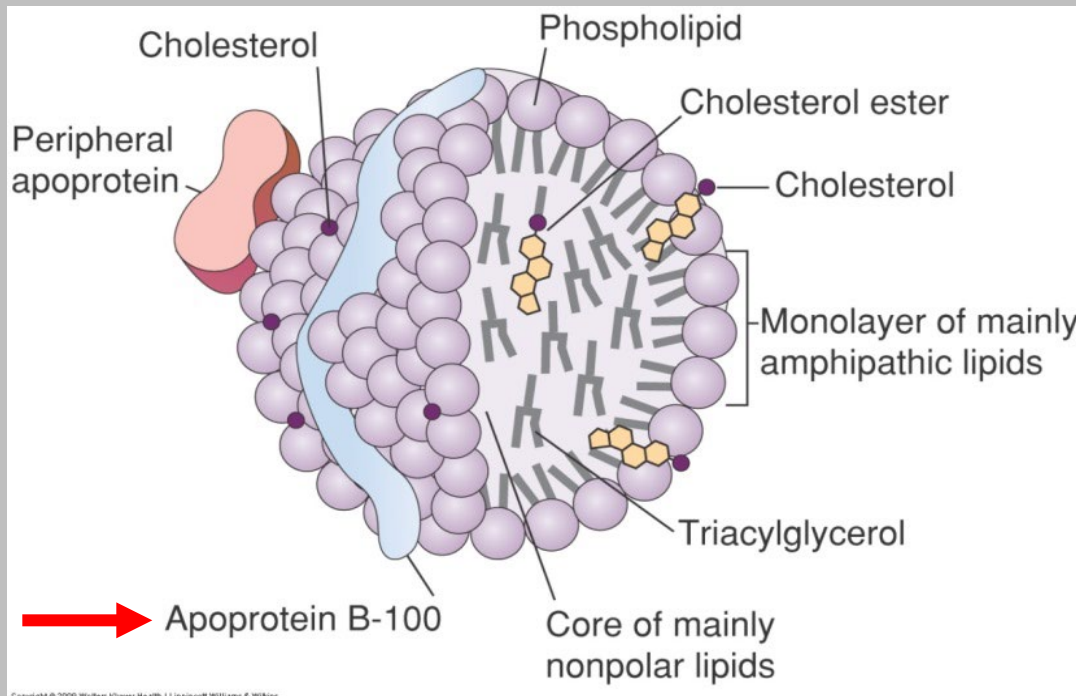
**Colipase:

- 1) Anchors lipase at the lipid-aqueous interface
- 2) Restores activity to lipase in the presence of inhibitory substances like bile salts that bind the micelles.

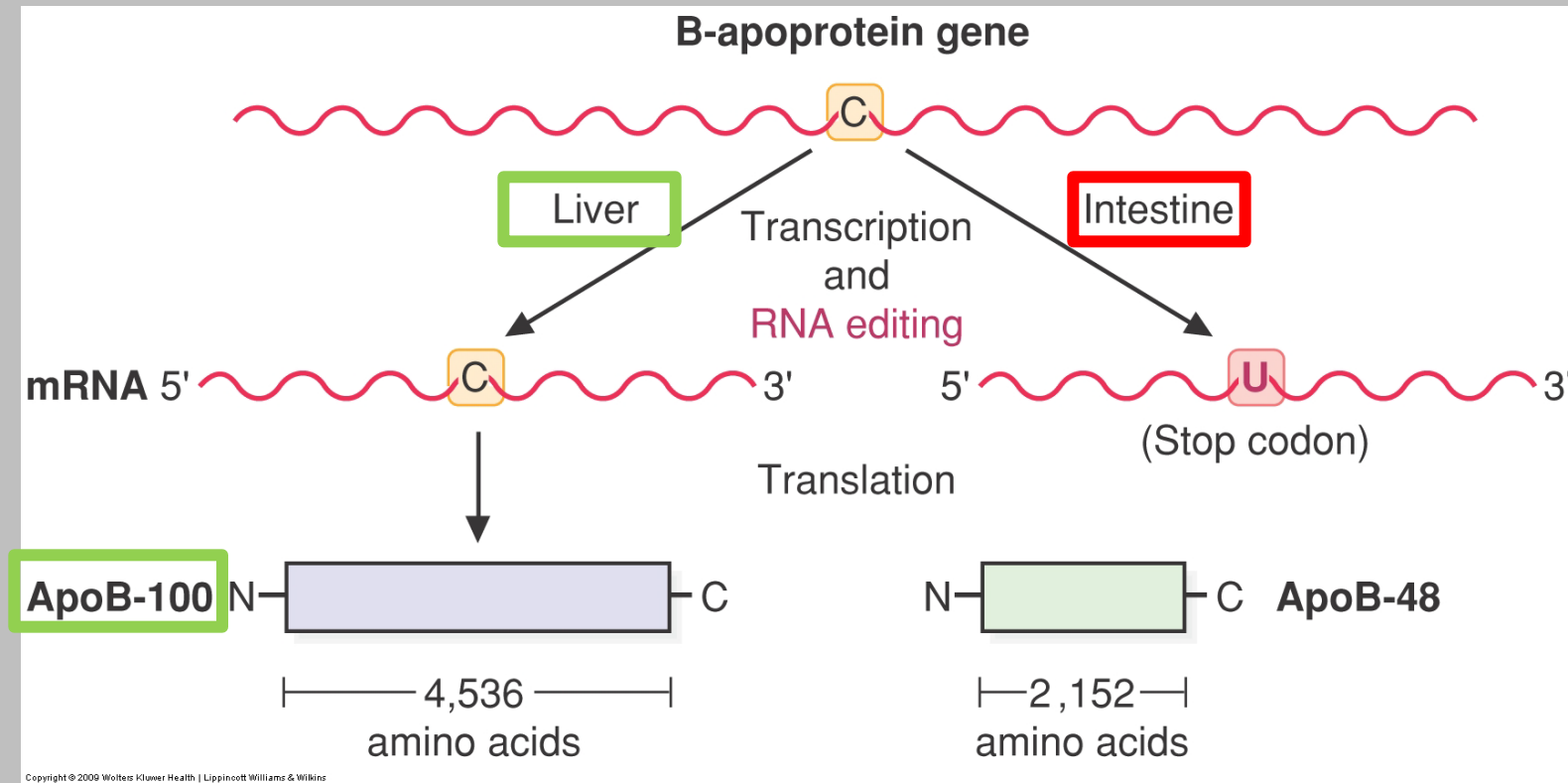
Transport of endogenous lipids – VLDL, IDL, LDL, and HDL

Transport of **endogenous** lipids by VLDL-I

- **VLDL** is synthesized in the **liver**
 - Secreted into the **hepatic veins** as **nascent VLDL**
 - Accept **apoE** and **apoC-II** from **HDL** in blood and become **mature VLDL**
 - Function to carry the lipids to peripheral tissues: muscle (used as energy), adipose (storage) etc.
- **ApoB-100** functions as the structural apoprotein for **VLDL** (next slide)
- **Microsomal triacylglycerol transfer proteins** (MTTP, or **MTP**) load lipids onto ApoB-100 to form **VLDL**
- **Lipoprotein lipase** (**LPL**) also digests the TGs of **VLDL**



Transport of **endogenous** lipids by VLDL: **ApoB-100**



VLDL

IDL

LDL

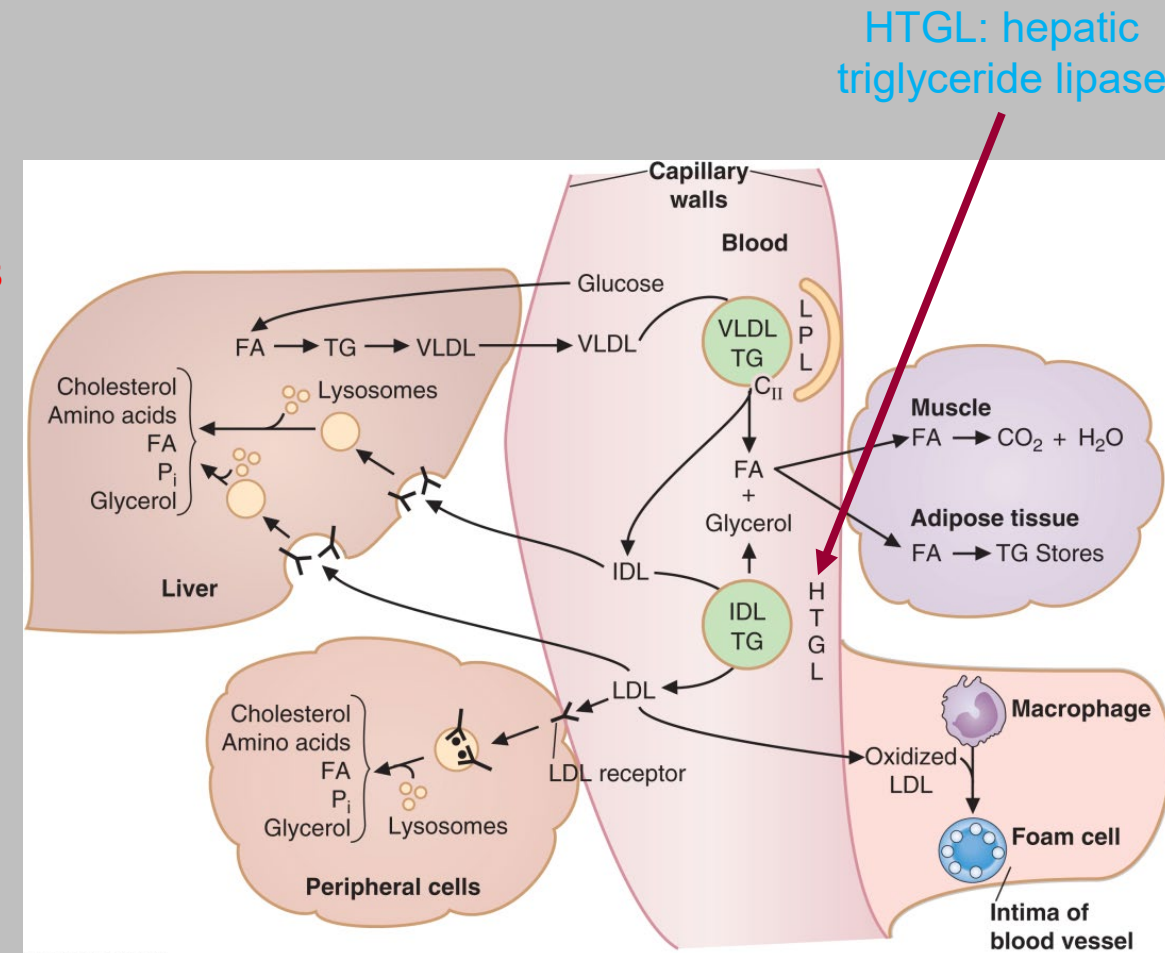
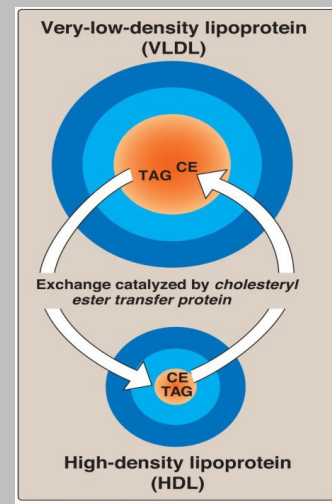
Chylomicron

Transport of **endogenous** lipids by VLDL-II

VLDL becomes **VLDL remnant** or **IDL** after losing TG

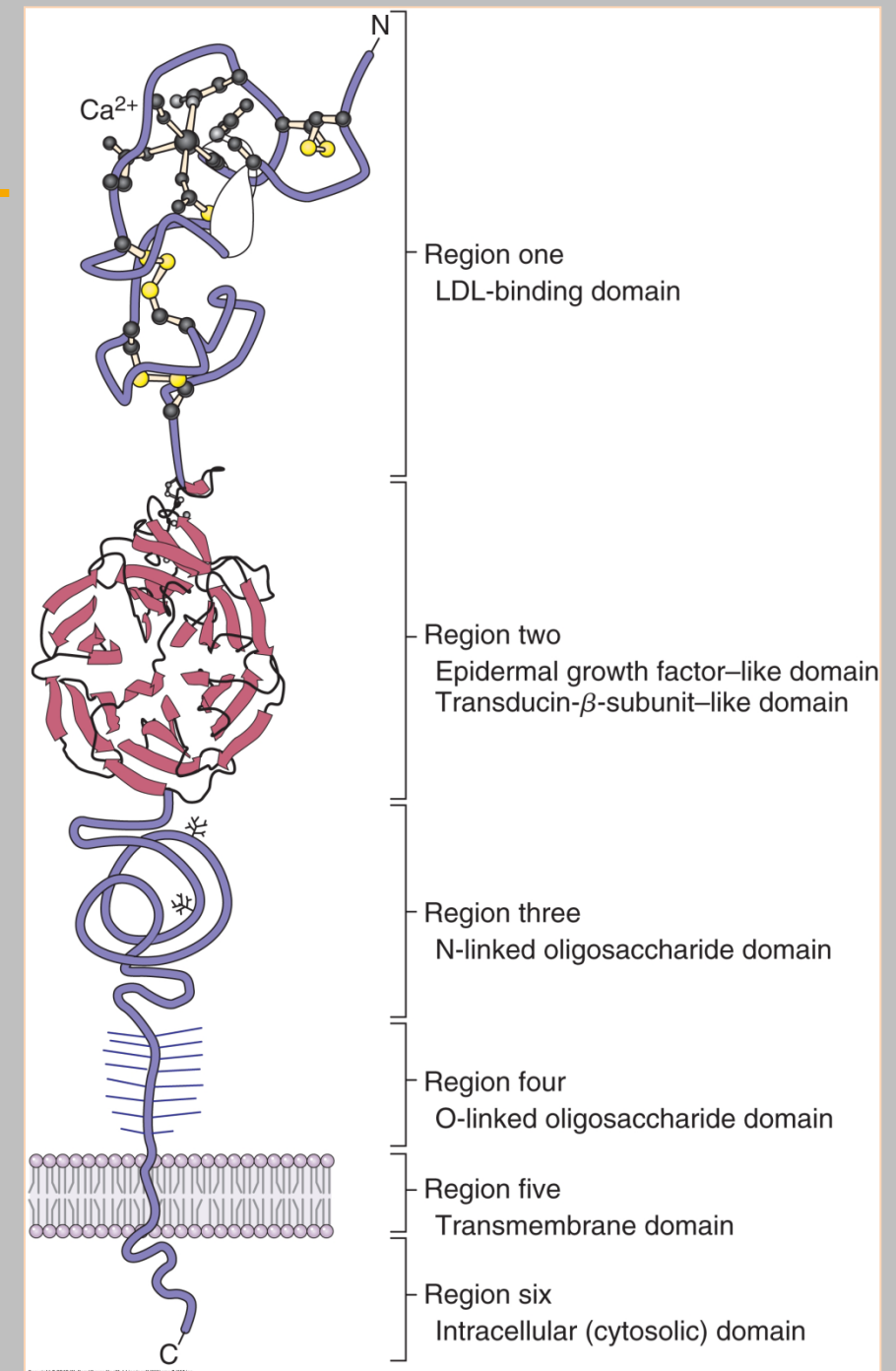
- **IDL** can be taken up by liver hepatocytes via **apoE receptor**
- **IDL** can also be further metabolized and become **LDL**
 - successive action of **LPL/HTGL** to further hydrolyze TGs
 - remaining **LDL** is rich of **cholesterol** (from **HDL**)

- **LDL** is taken up by liver or peripheral tissues
 - binds to the **LDL receptor** via **apoB-100**
 - **LDL** provides **cholesterol** to peripheral tissues
 - **high LDL levels can saturate the uptake by liver**
 - Excess and oxidized **LDL** taken up by **macrophages**
 - Forms **foam cells** then plaque - **atherosclerosis**



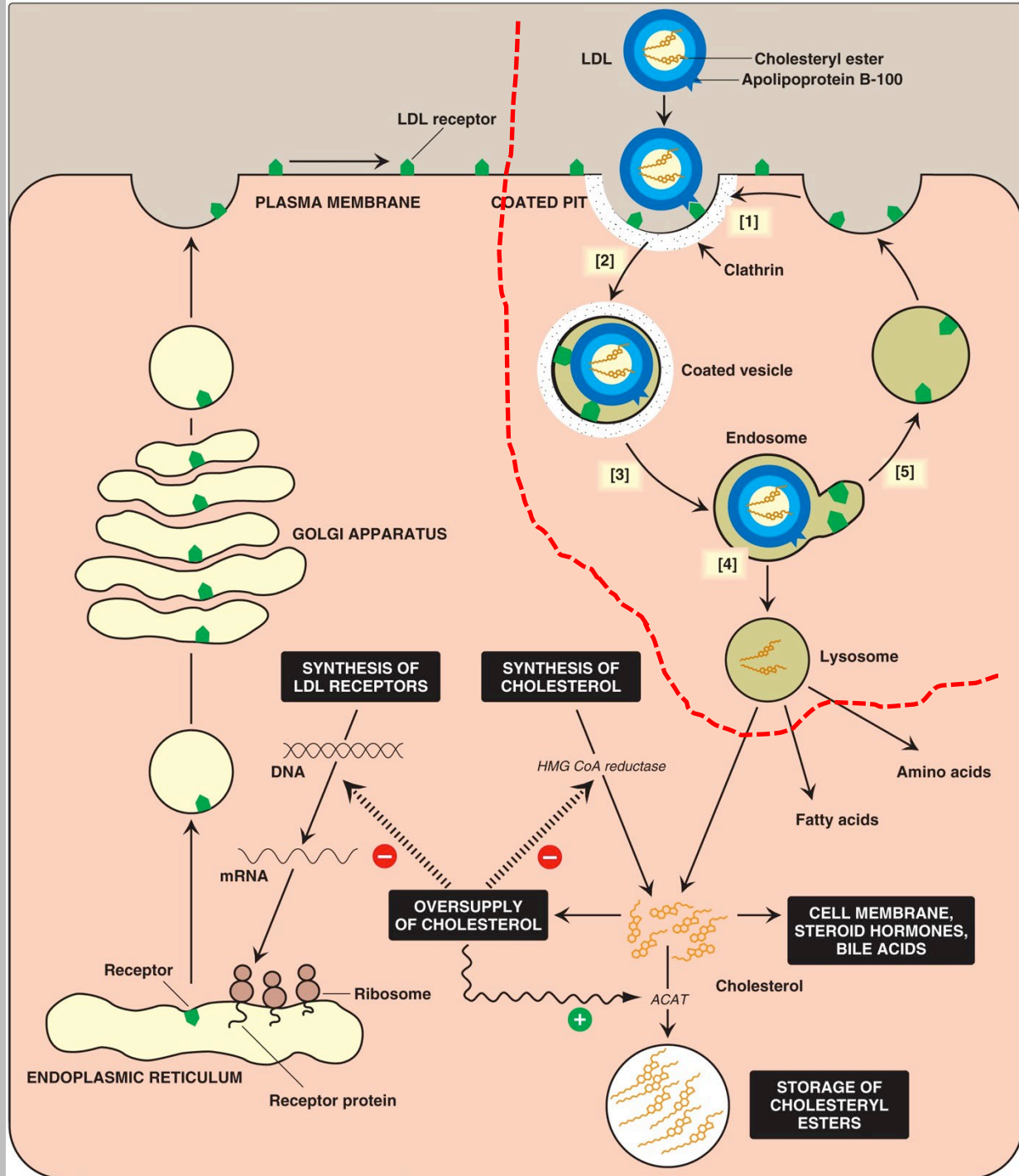
The LDL receptor

- The LDL receptor is found in many tissues.
- The LDL receptor is an **integral membrane protein** with six distinctive regions.
- The amino terminal domain is negatively charged and interacts with the **apoB-100** protein of LDL. The receptor can also bind **apoE** (found in VLDL and Chylomicron remnants).



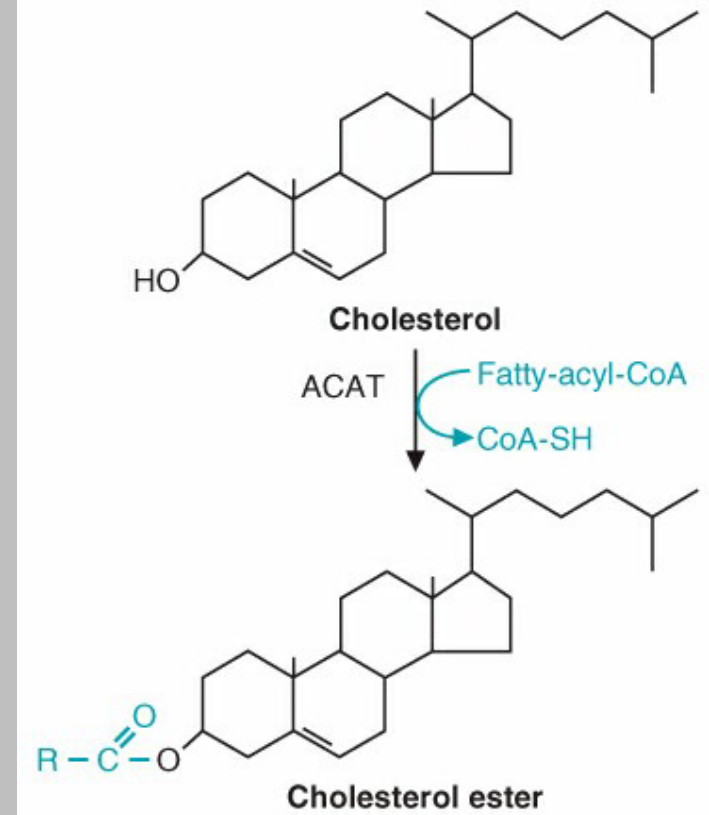
Cellular uptake and degradation of LDL

- Following LDL binding, the **receptor-LDL** complex is internalized via **endocytosis**. The endocytosed vesicles then fuse with **lysosomes** where the materials in the LDL are processed and/or re-used.
- The cholesteryl esters are hydrolyzed to liberate free cholesterol inside the cell.
- The receptors can be recycled back to the membrane.



Transport of **endogenous** lipids by VLDL - III

- Inside the cell, free cholesterol can be re-esterified by Acyl CoA:Cholesterol AcylTransferase (**ACAT**)
 - Esterification keeps cholesterol out of the cytoplasmic membrane
- LDL receptor synthesis can be down-regulated by cholesterol
 - Involves **SREBP** (as discussed in the cholesterol synthesis lecture)
- **Niemann-Pick type C disease** cannot release cholesterol from the lysosome:
 - Cholesterol builds up inside **lysosomes**
 - Cells (nerve cells) die → early death



HDL in reverse cholesterol transport

HDL in reverse cholesterol transport-I

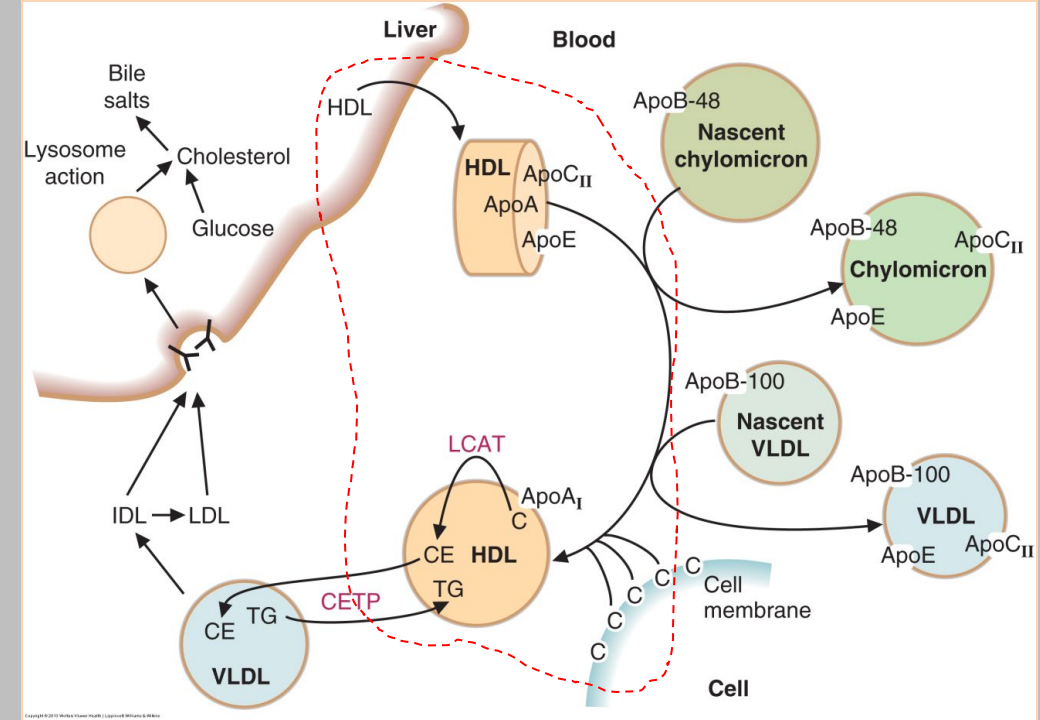
In contrast to other lipoproteins, the metabolism of HDL particles is not fully understood yet.

HDL can be synthesized in **liver** and **intestines**.

- **Nascent HDL** particle is **disk-shaped**, containing primarily **phospholipids**
 - Contains/transfers several apoproteins (**ApoA**, **ApoE**, and **ApoC-II**; others)
- **Mature HDL** also called **HDL3** (accumulating some cholesterol, next slide), **globular-shaped**
- High cholesterol HDL also called **HDL2** (atherogenic protective, next slide)

HDL plays a major role in **reverse cholesterol transport** (**vasculo-protective**)

- HDL picks up free cholesterol from the cell membranes of extrahepatic tissues and transfers them to liver or VLDL





Cells/tissues get rid of their cholesterol by **reverse cholesterol transport:**

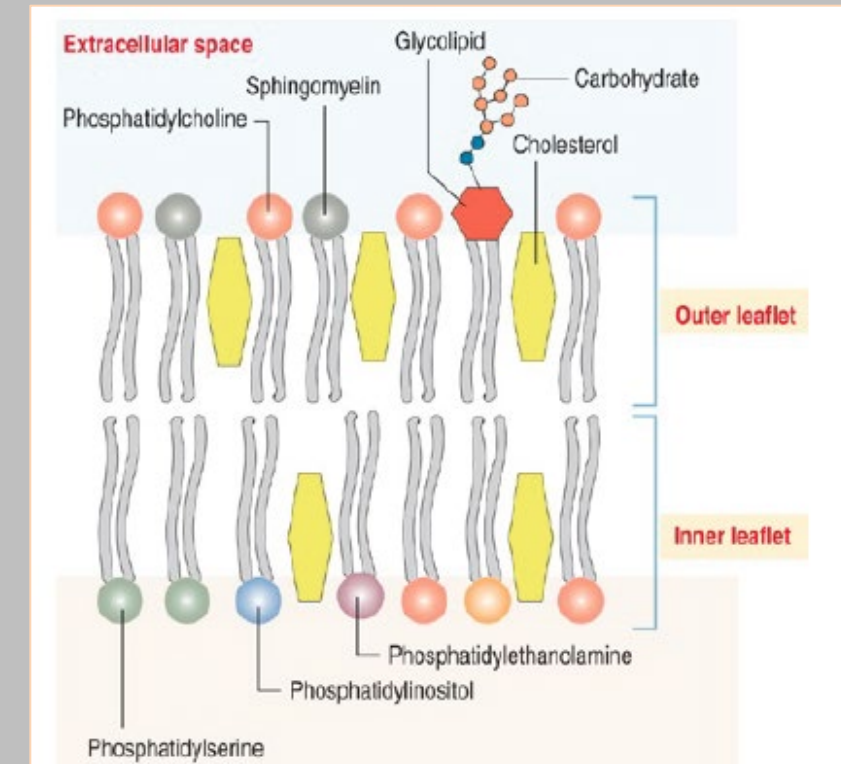
- **ABC1** (an ATPase, also known as ABCA1) flips cholesterol from the **inner leaflet** of plasma membrane to the **outer leaflet**
 - **Tangier disease**: ABCA1 deficiency → reduced HDL
- Cholesterol is then transferred to HDL

HDL picks up cholesterol from extrahepatic cells/tissues:

- Requires 2 enzymes:

LCAT (aka **PCAT-phosphatidylcholine**) = Lecithin: cholesterol acyl transferase (activated by **ApoA-I**)

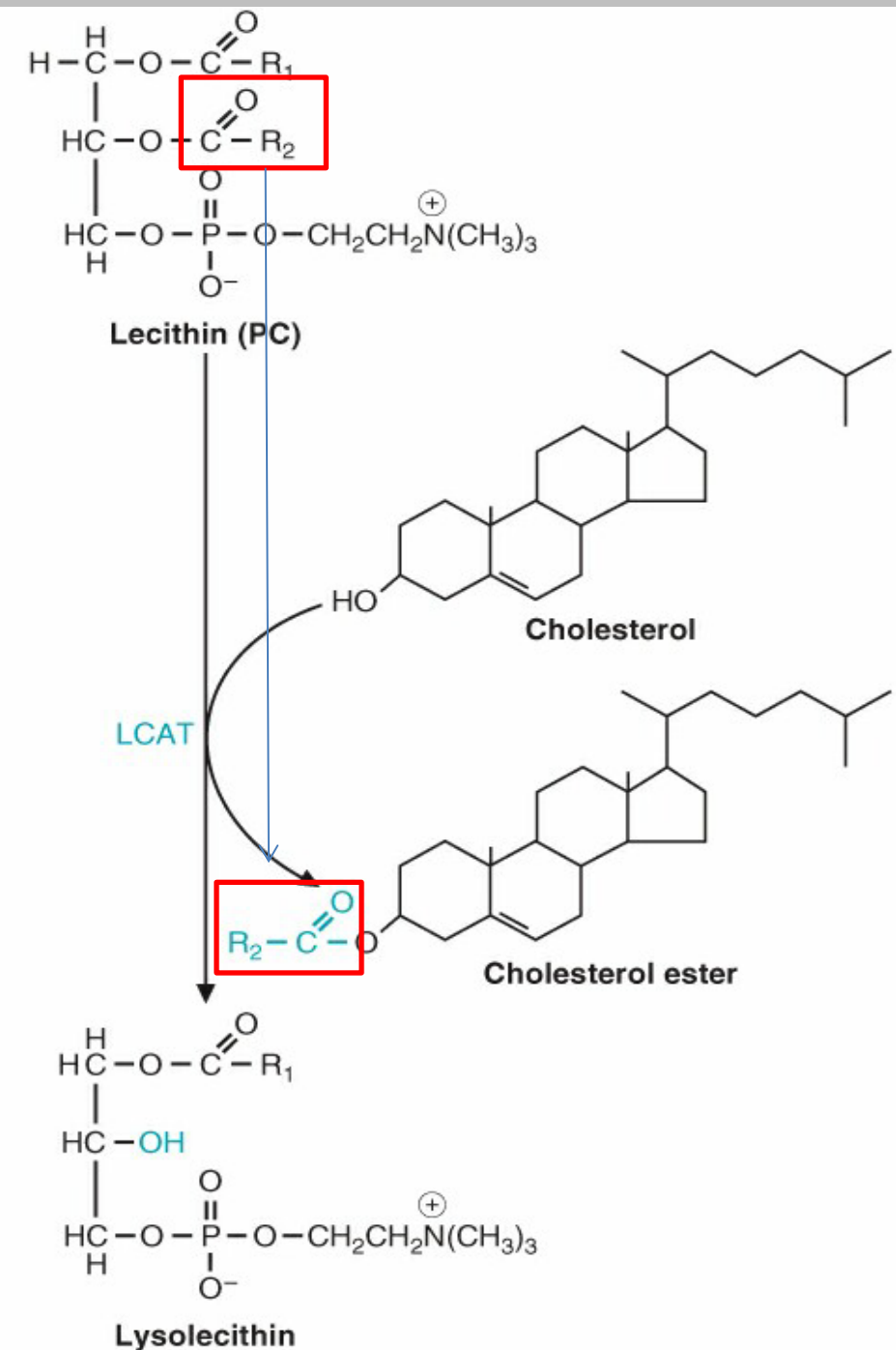
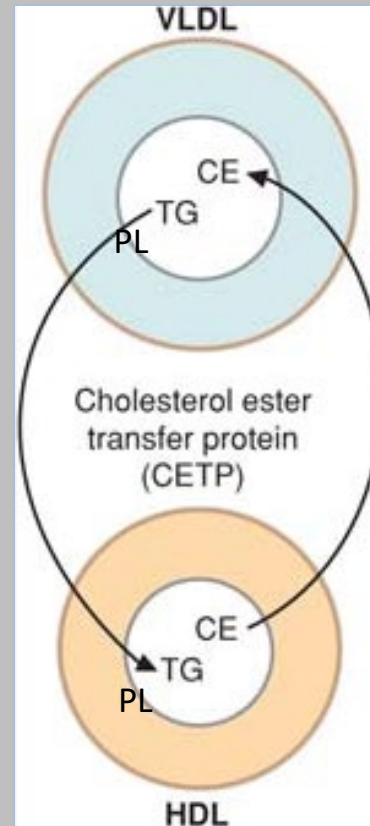
CETP = Cholesterol ester transfer protein



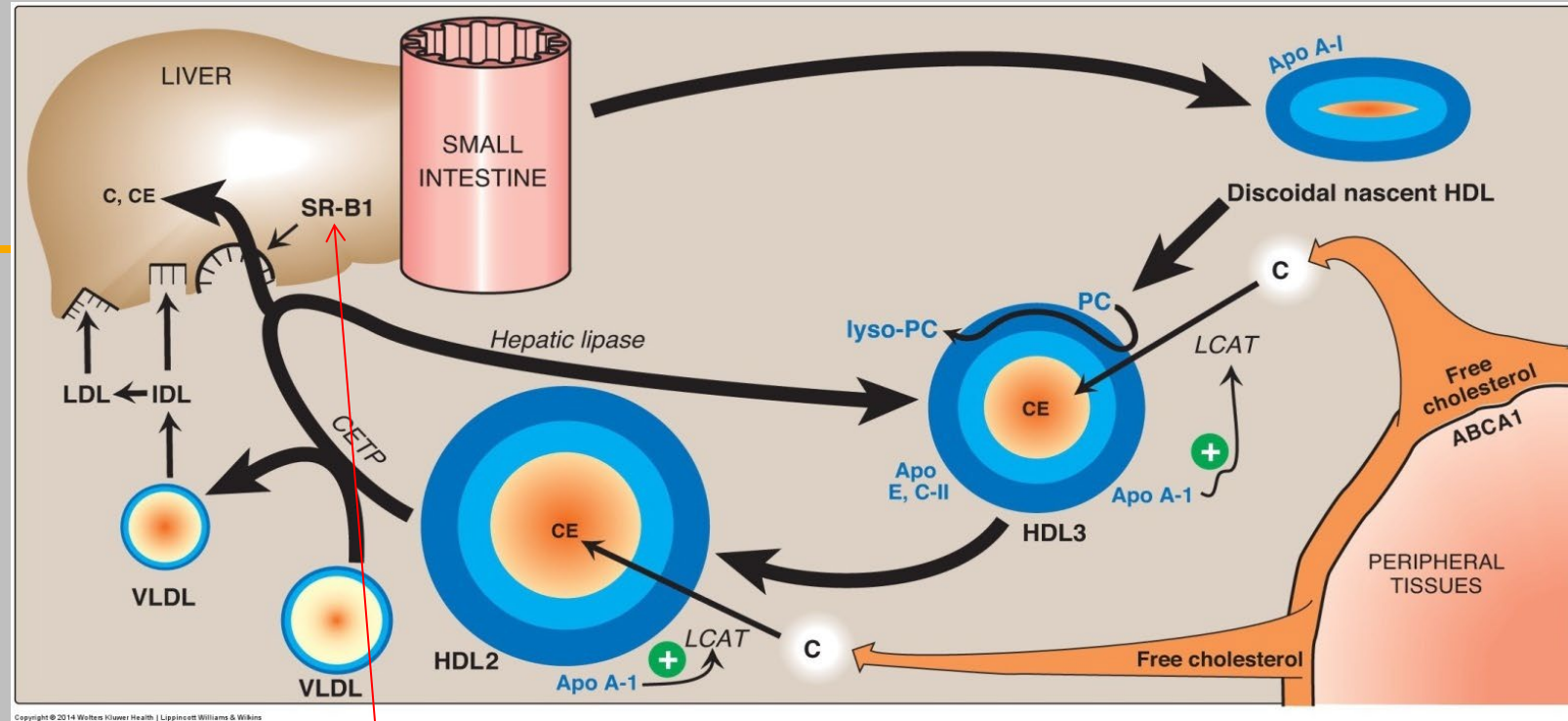
HDL in reverse cholesterol transport-III

HDL picks up cholesterol from extrahepatic cells/tissues requires 2 enzymes (**LCAT** and **CETP**):

- **LCAT** - lecithin cholesterol acyl transferase
 - Activated by **ApoA-I**
 - synthesizes cholesterol ester (CE)
 - FA comes from Phosphatidylcholine (lecithin)
 - **traps cholesterol in the HDL**
- **CETP** - cholesterol ester (CE) transfer proteins
 - transfers cholesterol ester to VLDL in exchange for triglyceride (TG) and phospholipid (PL)



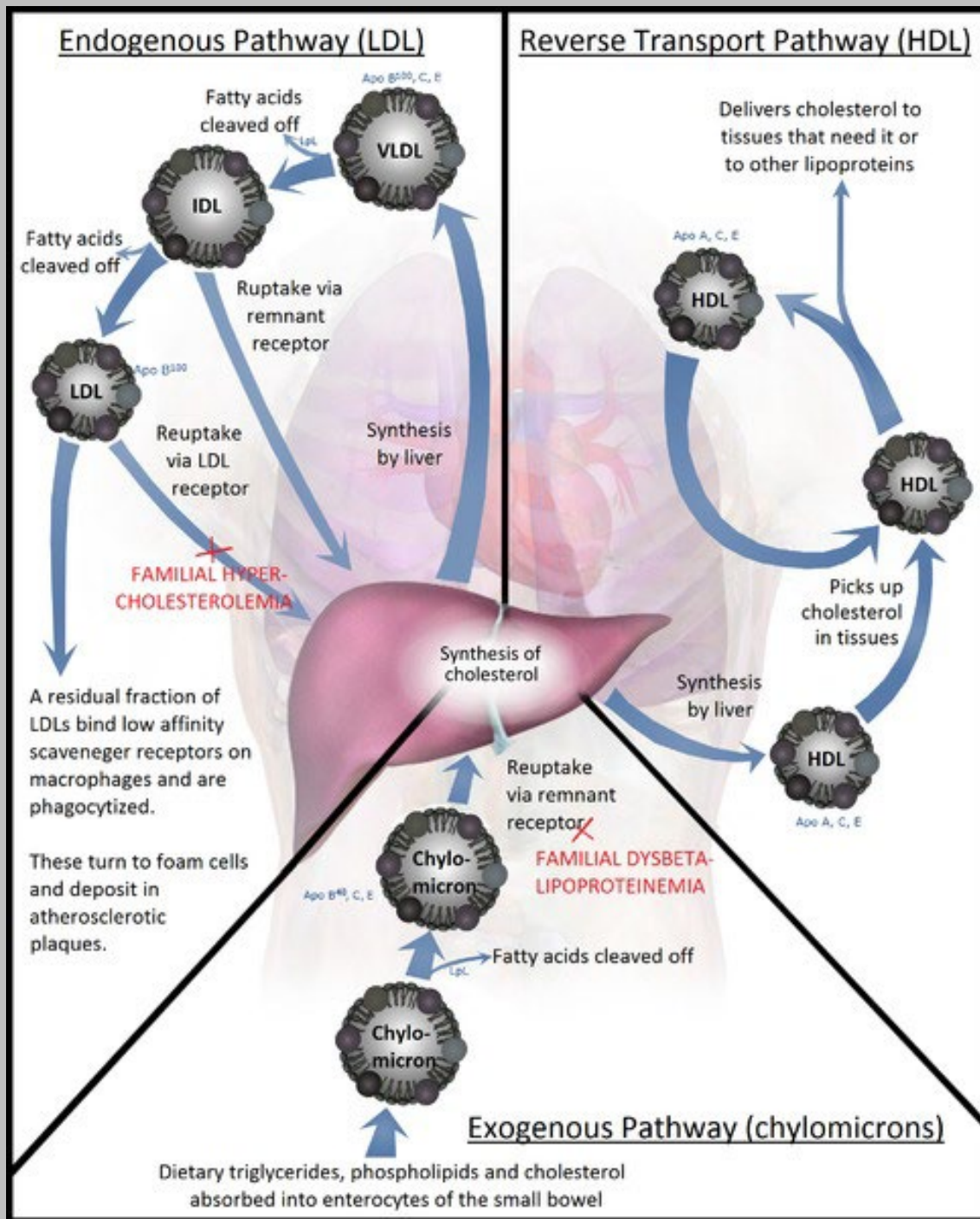
HDL in reverse cholesterol transport-IV



HDLs bind the **scavenger receptor (SR-B1)** on hepatocytes:

- Cholesterol (C) and cholesterol ester (CE) transferred into the liver cell
- **No endocytosis!!!** (unlike the LDL receptor)
- HDL dissociates and re-enters the circulation

Summary of plasma lipoproteins



Summary of plasma lipoproteins

Chylomicrons

- Produced in **intestinal** epithelial cells from **dietary fat**
- Carries lipids in blood

VLDL

- Produced in **liver**
- Carries lipids in blood

IDL

- Produced in blood (**remnant of VLDL** after TG digestion)
- Endocytosed by liver or converted to LDL

LDL

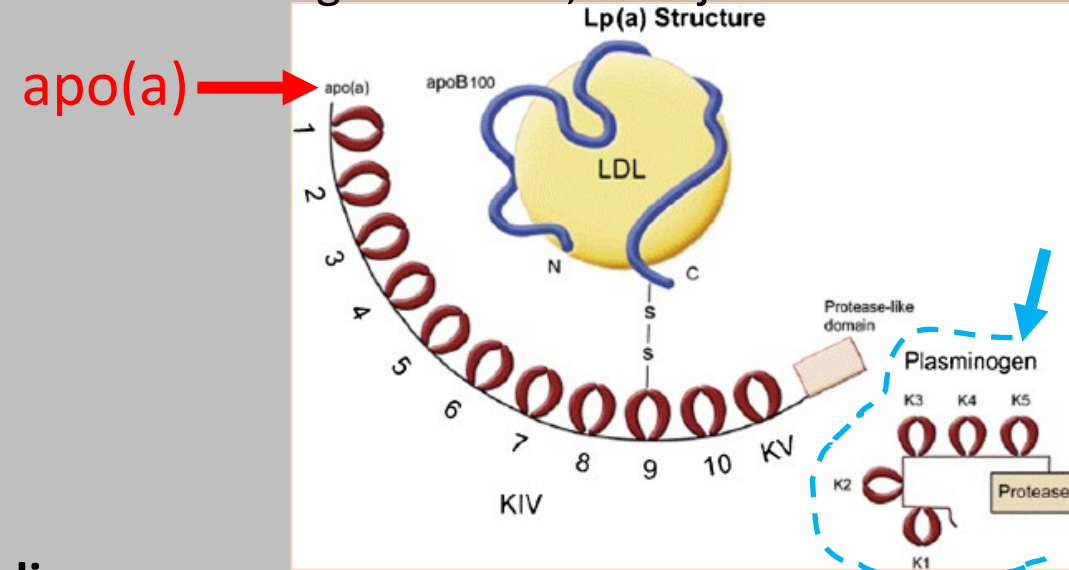
- Produced in blood (**remnant of IDL** after TG digestion; the end product of VLDL)
- Contain **high concentration of cholesterol and cholesterol ester**
- Endocytosed by liver and peripheral tissues via LDL receptors

HDL

- Produced in **liver** and **intestine**
- **Exchanges proteins and lipids** with other lipoproteins (e.g., chylomicrons, VLDL)
- Functions in the **return of cholesterol from peripheral tissues to liver, Reverse Cholesterol Transport**

Lp(a) - one last clinically important lipoprotein

- Lipoprotein(a), or Lp(a), is **nearly identical to LDL**. A distinguishing feature of Lp(a) is the presence of **apoprotein(a)**, or **apo(a)** that is covalently linked to a Cys residue on **apoB-100**
- **apo(a)** is structurally similar to **plasminogen** (a precursor of plasmin that degrades fibrin, a major component of blood clots.)

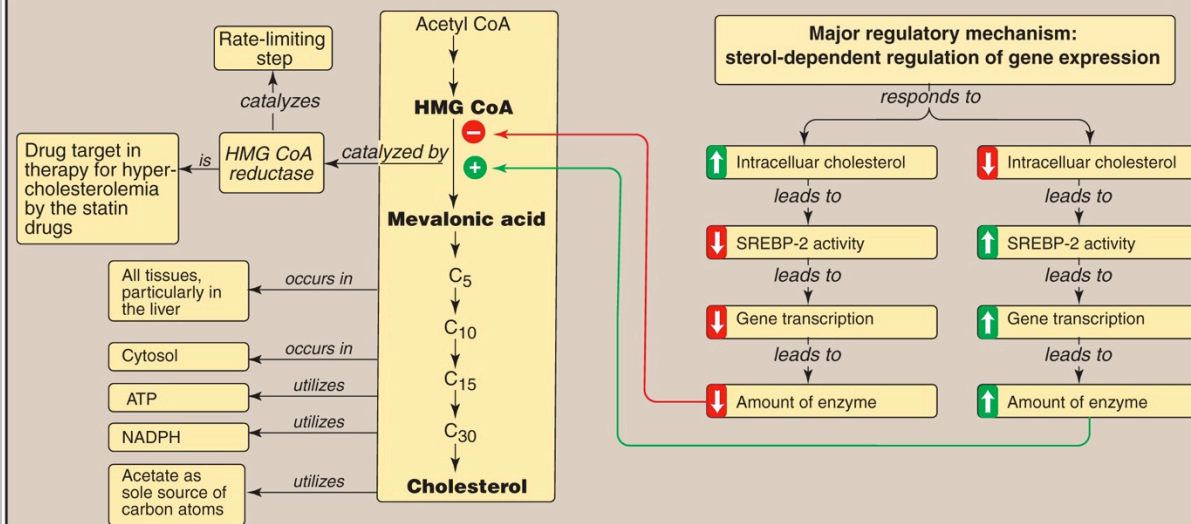


- High levels associated with increased risk of coronary heart disease
 - Levels mostly genetically controlled
 - High dietary **trans fats** are associated with increased level of Lp(a)
- The physiological function of Lp(a) remains unknown. One **hypothesis** is that elevated level of Lp(a) **slows the breakdown of blood clots** that trigger heart attack because it competes with **plasminogen** for binding of fibrin
- **Niacin** (vit. B3) reduces Lp(a), as well as LDL-cholesterol and TAG, and raises HDL

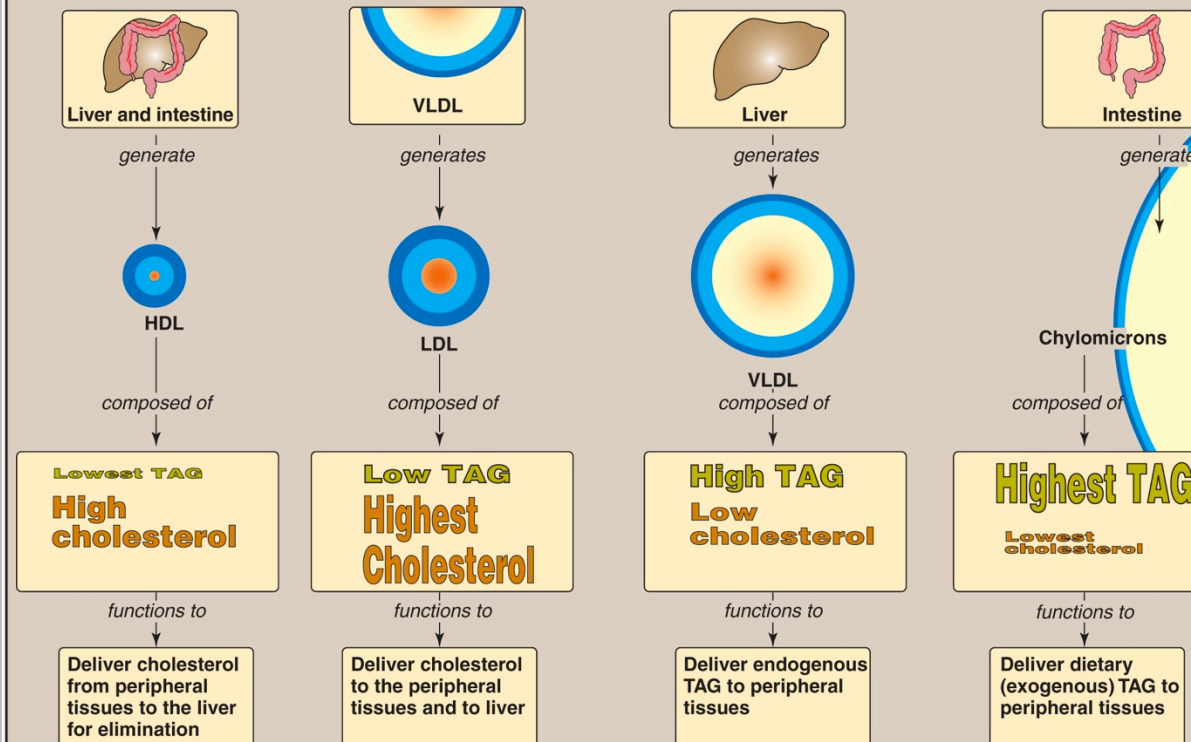
Summary

- **Key concepts:** chylomicrons, VLDL, IDL, LDL, HDL, Lp(a), apoproteins, MTP, Lipoprotein lipase (LPL), Reverse Cholesterol Transport
- Synthesis, maturation and fate of chylomicrons, VLDL, IDL, LDL, HDL
- Biological functions of chylomicrons, VLDL, IDL, LDL, HDL
- **Clinical correlations:** disease symptoms, biochemical basis, and treatment (Abetalipoproteinemia, fat malabsorption, Steatorrhea, Niemann-Pick type C disease, Tangier disease)

Cholesterol synthesis



Lipoproteins



Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>